EXPERIMENT :-5

PROGRAM CODES :-

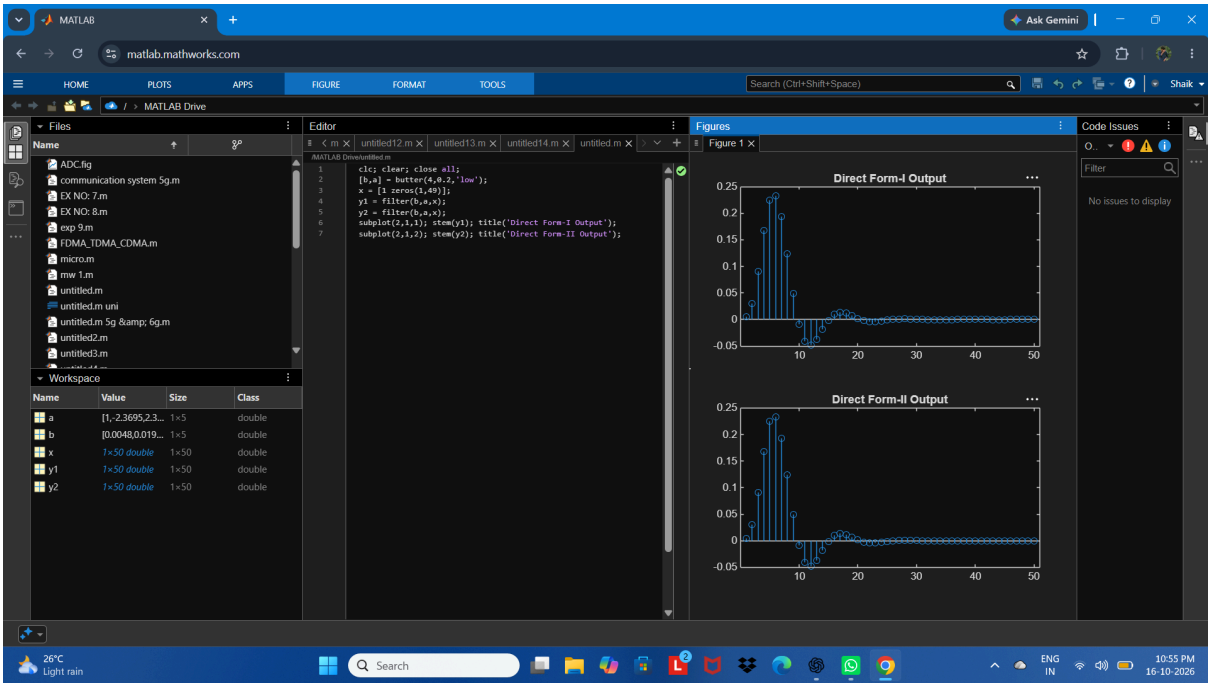
```
clc; clear; close all;
[b,a] = butter(4,0.2,'low');
x = [1 zeros(1,49)];
y1 = filter(b,a,x);
y2 = filter(b,a,x);
subplot(2,1,1); stem(y1); title('Direct Form-I Output');
subplot(2,1,2); stem(y2); title('Direct Form-II Output');
```

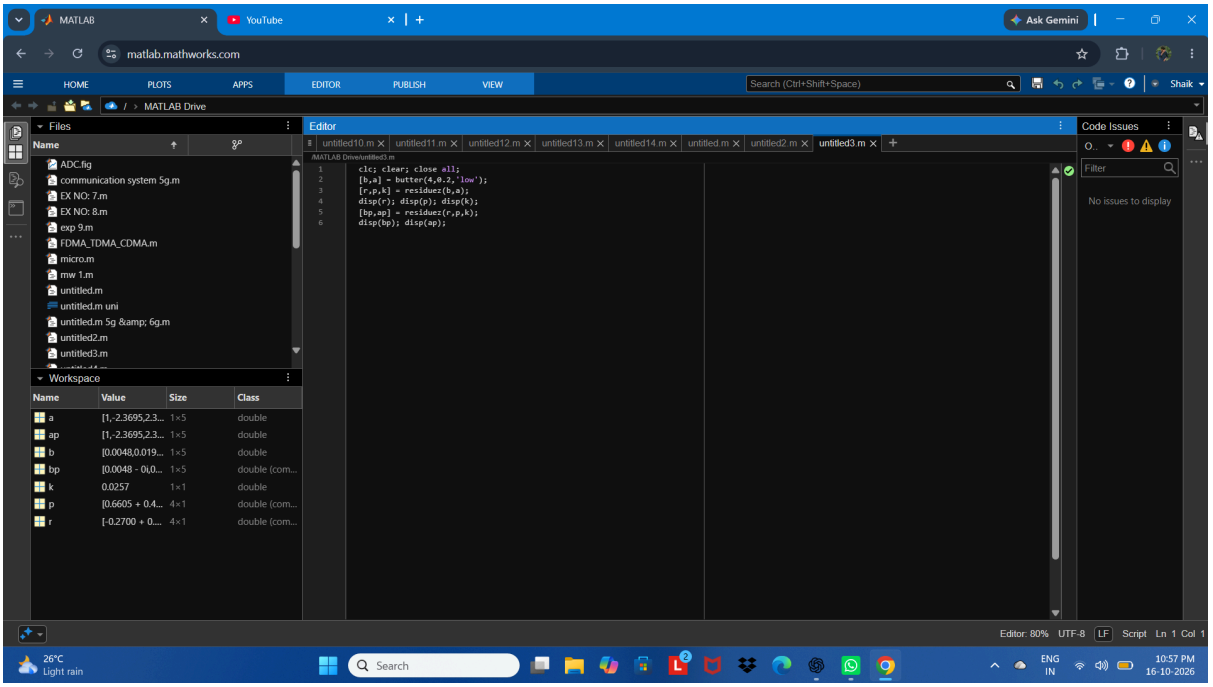
```
clc; clear; close all;
[b,a] = butter(4,0.2,'low');
[sos,g] = tf2sos(b,a);
disp(sos); disp(g);
x = [1 zeros(1,49)];
y = g* x;
for k = 1:size(sos,1)
    y = filter(sos(k,1:3),sos(k,4:6),y);
end
stem(y); title('Cascade Realization');
```

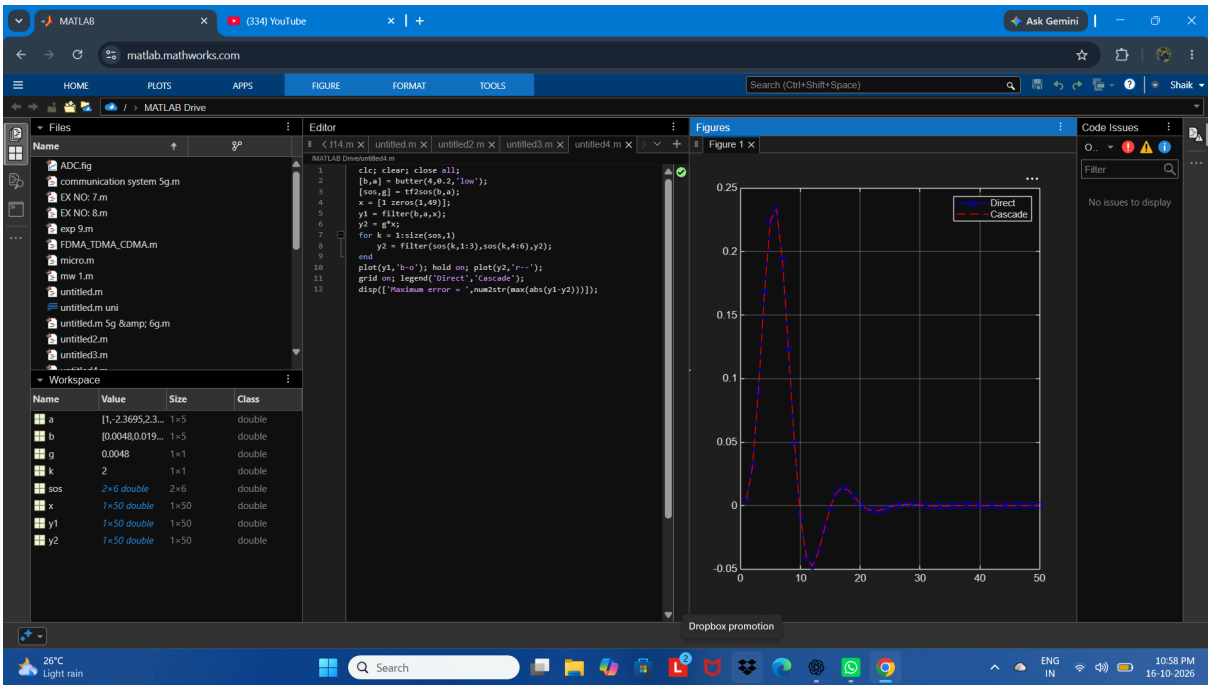
```
clc; clear; close all;
[b,a] = butter(4,0.2,'low');
[r,p,k] = residuez(b,a);
disp(r); disp(p); disp(k);
[bp,ap] = residuez(r,p,k);
disp(bp); disp(ap);
```

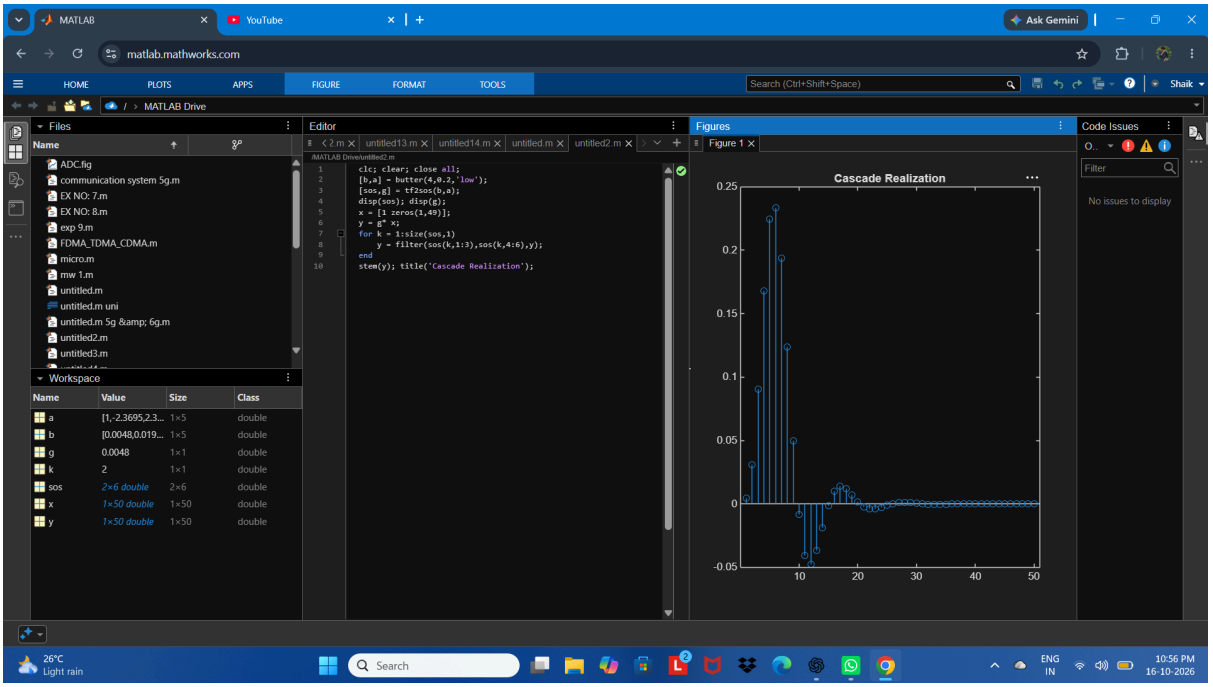
```
clc; clear; close all;
[b,a] = butter(4,0.2,'low');
[sos,g] = tf2sos(b,a);
x = [1 zeros(1,49)];
y1 = filter(b,a,x);
y2 = g*x;
for k = 1:size(sos,1)
    y2 = filter(sos(k,1:3),sos(k,4:6),y2);
end
plot(y1,'b-o'); hold on; plot(y2,'r--');
grid on; legend('Direct','Cascade');
disp(['Maximum error = ',num2str(max(abs(y1-y2)))]);
```

OUTPUTS:-




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MATLAB Drive
Files
Name
ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name Value Size Class
a [1,-2.3695,2.3... 1×5 double
ap [1,-2.3695,2.3... 1×5 double
b [0.0048,0.019... 1×5 double
bp [0.0048 - 0i,0... 1×5 double (com...
k 0.0257 1×1 double
p [0.6605 + 0.4... 4×1 double (com...
r [-0.2700 + 0.... 4×1 double (com...
Editor
untitled10.m
untitled11.m
untitled12.m
untitled13.m
untitled14.m
untitled.m
untitled2.m
untitled3.m
clc; clear; close all;
[b,a] = butter(4,0.2,'low');
[r,p,k] = residuez(b,a);
disp(r); disp(p); disp(k);
[bp,ap] = residuez(r,p,k);
disp(bp); disp(ap);
Code Issues
No issues to display
Editor: 80% UTF-8 LF Script Ln 1 Col 1
26°C
Light rain
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16-10-2026


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Files
Name
ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name Value Size Class
a [1,-2.3695,2.3... 1×5 double
b [0.0048,0.019... 1×5 double
g 0.0048 1×1 double
k 2 1×1 double
sos 2×6 double 2×6 double
x 1×50 double 1×50 double
y1 1×50 double 1×50 double
y2 1×50 double 1×50 double
Editor
f14.m
untitled.m
untitled2.m
untitled3.m
untitled4.m
clc; clear; close all;
[b,a] = butter(4,0.2,'low');
[sos,g] = tf2sos(b,a);
x = [1 zeros(1,49)];
y1 = filter(b,a,x);
y2 = g*x;
for k = 1:size(sos,1)
y2 = filter(sos(k,1:3),sos(k,4:6),y2);
end
plot(y1,'b-o'); hold on; plot(y2,'r--');
grid on; legend('Direct','Cascade');
disp(['Maximum error = ',num2str(max(abs(y1-y2)))]);
Figures
Figure 1
Direct
Cascade
0.25
0.2
0.15
0.1
0.05
0
-0.05
0
10
20
30
40
50
Code Issues
No issues to display
Dropbox promotion
26°C
Light rain
Search
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16-10-2026


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Name
ADC.fig
communication system 5g.m
EX NO: 7.m
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exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name
Value
Size
Class
a
[1,-2.3695,2.3...
1×5
double
b
[0.0048,0.019...
1×5
double
g
0.0048
1×1
double
k
2
1×1
double
sos
2×6 double
2×6
double
x
1×50 double
1×50
double
y
1×50 double
1×50
double
Editor
untitled13.m
untitled14.m
untitled.m
untitled2.m
clc; clear; close all;
[b,a] = butter(4,0.2,'low');
[sos,g] = tf2sos(b,a);
disp(sos); disp(g);
x = [1 zeros(1,49)];
y = g* x;
for k = 1:size(sos,1)
y = filter(sos(k,1:3),sos(k,4:6),y);
end
stem(y); title('Cascade Realization');
Figures
Figure 1
Cascade Realization
0.25
0.2
0.15
0.1
0.05
0
-0.05
10
20
30
40
50
Code Issues
Filter
No issues to display
26°C
Light rain
Search
ENG
IN
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